

Microtubule-dependent Oligomerization of Tau

IMPLICATIONS FOR PHYSIOLOGICAL TAU FUNCTION AND TAUOPATHIES*

Received for publication, May 18, 2003

Published, JBC Papers in Press, June 12, 2003, DOI 10.1074/jbc.M305207200

Victoria Makrides^{‡§}, Ting E. Shen[‡], Rajinder Bhatia[‡], Bettye L. Smith[‡], Julian Thimm[‡],
Ratneshwar Lal[‡], and Stuart C. Feinstein^{‡§¶}

From the [‡]Neuroscience Research Institute and [§]Department of Molecular, Cellular and Developmental Biology,
University of California, Santa Barbara, California 93106

The accumulation of abnormal tau filaments is a pathological hallmark of many neurodegenerative diseases. In 1998, genetic analyses revealed a direct linkage between structural and regulatory mutations in the tau gene and the neurodegenerative disease, frontotemporal dementia with parkinsonism linked to chromosome 17 (FTDP-17). Importantly, the FTDP-17 phenotype is transmitted in a dominant rather than a recessive manner. However, the underlying molecular mechanisms causing disease remain uncertain. The most common molecular mechanism generating dominant phenotypes is the loss of function of a multimeric complex containing both mutant and wild-type subunits. Therefore, we sought to determine whether tau might normally function as a multimer. We co-incubated ³⁵S-radiolabeled tau and biotinylated tau with taxol stabilized microtubules, at very low molar ratios of tau to tubulin. Subsequent covalent cross-linking followed by affinity-precipitation of the biotinylated tau revealed the formation of microtubule-dependent tau oligomers. We next used atomic force microscopy to independently assess this conclusion. Our results are consistent with the hypothesis that tau forms oligomers upon binding to microtubules. In addition to providing insights into normal tau action, our findings lead us to propose that one mechanism by which mutations in tau may cause cell death is through the formation of tau complexes containing mutant tau molecules in association with wild-type tau. These wild-type-mutant tau complexes may possess altered biological and/or biophysical properties that promote onset of the FTDP-17 phenotype, including neuronal cell death by either altering normal tau-mediated regulation of microtubule-dependent cellular functions and/or promoting the formation of pathological tau aggregates.

The microtubule-associated protein tau, localized predominantly in the cell bodies and axons of neuronal cells, is necessary for the establishment of neuronal cell polarity and axon outgrowth, for axonal transport, and the maintenance of axonal morphology (for example, see Refs. 1–6). Tau is also expressed in glial cells (7), although its role (8) there are less defined.

Mechanistically, tau is well known to stimulate MT¹ polymerization, to stabilize MTs, and to modulate MT dynamics (9–12). Since MT dynamics must be tightly regulated for cells to function and remain viable (*e.g.* see Refs. 13 and 14), it follows that tau action must also be finely regulated in cells.

Alternatively, abnormal tau behavior is often associated with neurodegenerative diseases. In Alzheimer's disease, FTDP-17 and a large number of additional "tauopathies," necrotic neurons possess abnormal pathological fibers composed primarily of hyperphosphorylated and dysfunctional tau (for a recent review, see Ref. 15). Until recently, the relationship between tau and these various diseases was only correlative. However, in 1998, several groups reported a direct genetic linkage between mutations in the tau gene and FTDP-17, a group of related neurodegenerative conditions characterized by neuronal cell death and dementia accompanied by abnormal tau fiber pathology (16–19). Although some of the tau mutations are structural and others are regulatory, all exhibit dominant phenotypes. Thus, both dysfunction and misregulation of tau are causally related to neuronal cell death, neurodegenerative disease, and dementia.

While there is only a single tau gene, alternative splicing of tau mRNA produces two classes of tau proteins known as "3-repeat tau" and "4-repeat tau" (20, 21). Previous work has shown that 4-repeat tau binds to MTs ~3-fold stronger than 3-repeat tau (22–24). It also assembles MTs more effectively than 3-repeat tau (25, 26). Finally, 4-repeat tau is a significantly stronger stabilizer of microtubule dynamics, both *in vitro* (12) and in cells.²

The FTDP-17 mutations introduce amino acid substitutions in or near to the MT binding domain or alter the pattern of tau RNA splicing without altering the primary sequence of the encoded tau protein (16–18, 27). These latter mutations result in an increase in the expression ratio of 4-repeat tau to 3-repeat tau in adult neurons. Although the genetic linkage between tau mutations and FTDP-17 establishes a clear cause-and-effect relationship between tau dysfunction/misregulation and neuronal cell death and dementia, the underlying molecular mechanism leading to dementia and neuronal cell death is not understood. In line with the growing number of dominantly inherited degenerative diseases involving abnormal protein folding (28), a number of investigators have suggested a "gain-of-a-toxic-function" mechanism in which the mutations confer

* This work was supported by United States Public Health Service Grants NS35010 (to S. C. F.) and GM056290 (to R. L.) and by the California Department of Health Services, Alzheimer's Disease Program (to R. L.). The costs of publication of this article were defrayed in part by the payment of page charges. This article must therefore be hereby marked "advertisement" in accordance with 18 U.S.C. Section 1734 solely to indicate this fact.

¶ To whom correspondence should be addressed: Neuroscience Research Inst. and Dept. of Molecular, Cellular, and Developmental Biology, University of California, Santa Barbara, CA 93106. Tel./Fax: 805-893-2659; E-mail: feinstein@lifesci.ucsb.edu.

¹ The abbreviations used are: MT, microtubule; AFM, atomic force microscopy; EDC, 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide; FTDP-17, frontotemporal dementia with parkinsonism linked to chromosome 17; HPLC, high performance liquid chromatography; MAP, microtubule-associated protein; Pipes, 1,4-piperazinediethanesulfonic acid.

² J. M. Bunker, L. Wilson, M. A. Jordan, and S. C. Feinstein, manuscript in preparation.

upon tau an increased probability of forming abnormal tau fibers, which are in turn toxic to the cells (29, 30). However, while this is a plausible model for FTDP-17 mutations that alter the tau amino acid sequence, it is less obvious what toxic function might be acquired by the tau RNA splicing mutations in which only the 3-repeat to 4-repeat ratio of otherwise wild-type proteins is affected.

An alternative to the gain-of-a-toxic function model is a loss-of-function model. Indeed, the most common molecular mechanism underlying dominant mutations is poisoning of, or otherwise altering the properties of, an oligomeric complex that is essential for cell viability. In the case of tau, if proper tau function were to require oligomerization, perhaps oligomers containing mutant tau polypeptides harboring amino acid substitutions or an altered ratio of wild-type tau isoforms alter function sufficiently to cause cell death.

Here, we employed two independent assays to test the hypothesis that tau oligomerization may be a part of normal tau action. A biochemical assay demonstrates that, even at very low tau:tubulin molar ratios, tau-tau dimers (and potentially larger oligomers) form in a microtubule dependent manner. Second, AFM visualization of tau alone, tau and microtubules at a low tau:tubulin molar ratio, and tau and MTs at a saturating tau:tubulin ratio support the conclusion that tau oligomerizes in a microtubule dependent manner. These tau oligomers are likely to be important for tau action during normal neuronal development; they also present a molecular mechanism for the tau-mediated cell death in neurodegeneration.

MATERIALS AND METHODS

Protein Reagents—Full-length rat 4R2N tau cDNA cloned into pET23a was *in vitro* transcribed and translated (Promega TNT wheat germ lysate kit) using [³⁵S]methionine (PerkinElmer Life Sciences). Product yields were determined by percent specific incorporation of radioactivity. Recombinant tau was purified by phosphocellulose chromatography and HPLC (31). Recombinant tau concentrations were determined by comparing to known tau mass standards (established by mass spectroscopy) by quantitative immunoblotting using a tau polyclonal affinity purified antibody directed against full-length 3-repeat tau (pan-tau, a kind gift of Monte Radeke) followed by ECL detection (Pierce).

Tau was biotinylated according to the manufacturer's protocol (Pierce) with a 7.5:1 molar ratio of biotin to tau. Briefly a 10 mg/ml working stock of NHS-biotin (Pierce) was prepared in 50 mM Sodium Bicarbonate (pH 8.0) buffer immediately prior to use. Biotin was added to 200 μ g of a 5 mg/ml solution of tau (4.43×10^{-6} M) at a final concentration of 6.645×10^{-5} M biotin. The solution was gently mixed, incubated for 2 h on ice, and free biotin was removed by desalting through a CS-20 column (Princeton Separations, Inc.).

Microtubule Polymerization—Taxol stabilized MTs were assembled from MAP-free phosphocellulose-purified bovine tubulin (a kind gift of Dr. Les Wilson and Herb Miller) by incubation at 37 °C in BRB-80 buffer (80 mM Pipes, 1 mM EGTA, 1 mM MgSO₄, pH 6.8) in the presence of 1 mM GTP and the stepwise addition of increasing concentrations of taxol (diluted in Me₂SO) to a final concentration of 10 μ M taxol (Sigma). MT dilutions were made in BRB-80, 1 mM GTP, 10 μ M taxol.

1-Ethyl-3-(3-dimethylaminopropyl)carbodiimide (EDC) Cross-linking Assays—Tau was incubated either with, or without, taxol-stabilized MTs for 30 min at 37 °C in BRB-80 buffer, 1 mM GTP, 10 μ M taxol. EDC (Sigma) was added at the final concentrations indicated (0.1–1.5 mM) in BRB-80 buffer, and the mixture was incubated for 30 min at room temperature. Cross-linking was terminated by the addition of 1 volume of 2 $\times$ protein sample buffer (100 mM Tris, pH 6.8, 1% SDS, 20% glycerol, 1.4 M β -mercaptoethanol) with or without the addition of 10 mM glycine (as indicated). Samples were fractionated by SDS-PAGE and either directly prepared for autoradiography or transferred to a membrane (nitrocellulose or polyvinylidene difluoride) for immunoblotting prior to autoradiography.

Microtubule Assembly Assay—Phosphocellulose-purified tubulin at a final concentration of 2 mg/ml in BRB-80 buffer supplemented with 1 mM GTP was added to varying concentrations of recombinant tau and incubated at 37 °C for 30 min, then gently layered over a prewarmed 50% sucrose cushion and ultracentrifuged at 100 kg for 15 min at 25 °C.

Pellets and supernatants were recovered and fractionated on SDS-PAGE, Coomassie-stained, and the amount of tubulin and tau in each fraction was determined by comparison to known mass standards.

Biotin Pull-down Assay—MT assembly was carried out by two rounds of sequential polymerization. Initially MT seeds were prepared by assembling 10% of the final volume of MAP-free phosphocellulose tubulin in BRB-80 buffer, 1 mM GTP, 10% Me₂SO, 10% glycerol at 37 °C for 30 min, followed by shearing. The remaining tubulin was added to the microtubule seed preparation and polymerized under the same buffer and temperature conditions as described previously. MTs were then incubated with 0.1 μ M recombinant, HPLC-purified, biotinylated tau for 30 min at 37 °C. EDC cross-linking was carried out as described above and cross-linking was quenched with the addition of a 10-fold molar excess of glycine and the sample denatured by treatment with 1% SDS and heating (two intervals of 2.5 min at 90 °C followed by 2.5 min at room temperature). The samples were then diluted into 10 volumes of binding buffer (20 mM Tris, pH 7.4, 150 mM NaCl) with 1% Triton X-100 and incubated on ice for 20 min. An excess of Softlink™ monomeric avidin resin (Pierce) was added to each reaction, and the samples were incubated 16 h at 4 °C with tumbling. The avidin beads were extensively washed in binding buffer with 0.1% Triton X-100, and biotinylated tau was eluted at room temperature with 1 $\times$ protein sample buffer.

Atomic Force Microscopy—AFM was performed using a Digital Instruments Multimode Nanoscope III AFM equipped with a D-scanner (15 μ m scan size). Tappingmode™ AFM was performed in fluid using a Tappingmode™ Liquid Cell (Digital Instruments) and oxide-sharpened silicon nitride tips (with a spring constant of 0.06 Newton/m (DNPS, Nanoprobe, Digital Instruments)).

For imaging of individual tau molecules, samples were diluted to 5 ng/ml in BRB-80 buffer containing 5 mM β -mercaptoethanol and incubated for 30 min at room temperature and then adsorbed onto freshly cleaved mica for 5 min at room temperature. Samples were rinsed and imaged in sterile filtered BRB-80 buffer. Taxol-stabilized MTs were prepared for imaging with a prepelleting step of 25 kg, for 6 min, at room temperature; pellets were then washed twice with BRB-80 buffer and gently resuspended in warm filtered BRB-80, 1 mM GTP, 10 μ M taxol. Prior to imaging, MT samples were fixed in 0.2% glutaraldehyde (8% stock, Ted Pella) for no less than 10 min to no more than 4 h. 20 μ l of the MT samples were then adsorbed onto 3-aminopropyltriethoxysilane (Sigma) coated mica (freshly prepared by 15 min of vapor deposition) for 15 s, rinsed twice with one ml of BRB-80 buffer, and imaged in sterile filtered buffer.

Unless otherwise noted, the AFM data were minimally filtered using a first order flatten and/or a first order plane fit function. Sample dimensions were determined from the topography images. The formula for determining the volume (*V*) of a partial sphere: $V = [\pi h (3r^2 + h^2)]/6$ was used to calculate the mean volume of tau monomers from the mean height (*h*) and apparent diameter (*2r*) as measured by AFM. To compensate for tip broadening effects, apparent protein diameters were determined at half the maximum height (32).

RESULTS

EDC Cross-linking of Tau Bound to Microtubules Leads to the Formation of Microtubule-dependent, Tau-containing Complexes—As a first test of the hypothesis that tau might homooligomerize under normal, non-pathological conditions, we used the zero-length cross-linker EDC to covalently cross-link MTs bound to equilibrium with 4-repeat tau. As seen in Fig. 1A, (lanes 1 and 4) radiolabeled 4-repeat tau migrates as a single major band on SDS-PAGE with an apparent molecular mass of ~65 kDa. Treatment of tau with 1.5 mM EDC (Fig. 1A, lanes 2 and 5) results in the formation of no new products. However, cross-linking of radiolabeled tau bound to taxol-stabilized MTs (~3 nM tau and 20 μ M MTs) leads to the generation of a series of radiolabeled bands between ~80–160 kDa and of greater than ~230 kDa (Fig. 1A, lanes 3 and 6). Control reactions in which the non-MT-binding protein luciferase (62 kDa) was substituted for tau exhibited no MT-dependent bands (Fig. 1B), demonstrating the specificity of the MT binding dependent cross-linking events for tau.

Tau Binding to Microtubules Leads to Direct Tau-Tau Association—The various MT dependent, cross-linked complexes observed in lanes 3 and 6 of Fig. 1A could be multimers com-

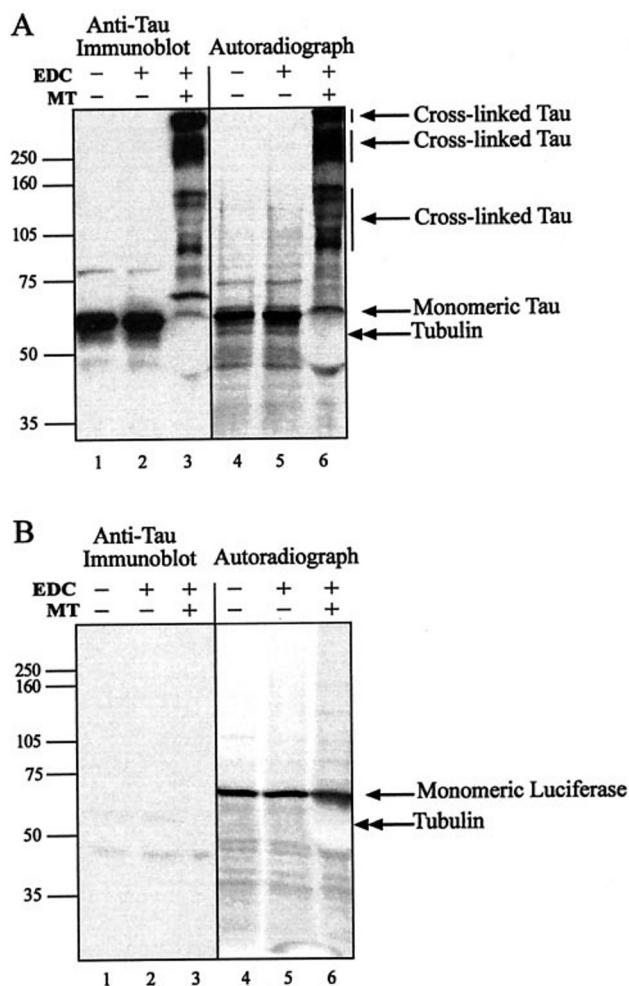


FIG. 1. EDC cross-linking of tau bound to MTs leads to the formation of MT-dependent tau containing complexes. For all panels, the left side is the immunoblot (probed with anti-pan-tau antibody), and the right side is the autoradiograph of the same gel. **A**, tau cross-linking by EDC as a function of MT binding. *In vitro* translated ^{35}S -labeled tau (~ 3 nM) was fractionated on 8% SDS-polyacrylamide with and without prior binding to MTs ($23 \mu\text{M}$) (as indicated) and/or with or without cross-linking (1.5 mM EDC as indicated). **B**, EDC does not cross-link the non-MT-binding protein, luciferase. The analogous experimental conditions to **A** are shown using *in vitro* translated ^{35}S -labeled luciferase (6 nM).

posed either solely of tau or contain both tau and tubulin. To test the hypothesis that MT binding might promote tau-tau intermolecular associations, we developed an assay in which we 1) co-incubated biotinylated tau, radiolabeled tau, and taxol-stabilized MTs, 2) cross-linked with EDC under conditions optimized to maximize the quantity of cross-linked complexes in the region of the gel predicted for tau-tau dimers, and 3) affinity-purified the biotinylated tau with avidin beads. The presence of radiolabeled tau co-precipitating with the biotinylated tau with an apparent size consistent with tau dimers would support the hypothesis that tau multimers form on MTs.

As controls, we first established that biotinylation of tau does not affect its functional capability, as assessed by its ability to assemble tubulin into MTs (data not shown). An additional control experiment demonstrated that biotinylated tau bound to MTs produces a similar collection of cross-linked complexes as radiolabeled tau (Fig. 2A, lanes 1 and 2). This same control experiment further demonstrates that no detectable tubulin-containing cross-linked products form using these low concentrations of EDC (Fig. 2A, lanes 5 and 6), and further, only

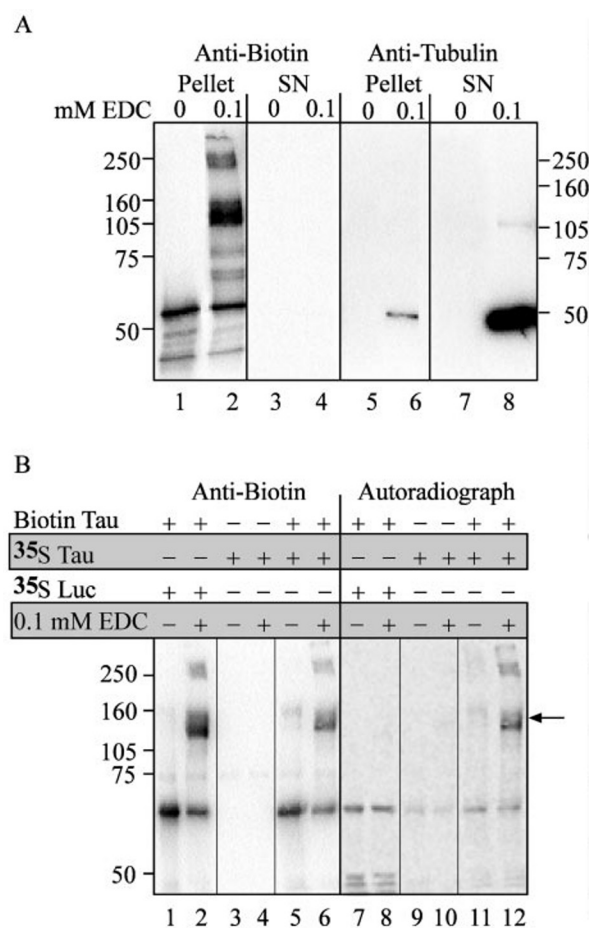


FIG. 2. Tau binding to MTs leads to direct tau-tau association. **A**, biotinylated tau is specifically and efficiently immunoprecipitated. $0.1 \mu\text{M}$ biotinylated tau was cross-linked to $20 \mu\text{M}$ MTs. Following avidin immunoprecipitation the presence of biotinylated tau versus tubulin in the pellets versus in the supernatants was compared by immunoblotting with streptavidin-horseradish peroxidase versus a tubulin-specific monoclonal antibody (DM1A, Sigma). **B**, comparison of the biotin immunoblot and the autoradiograph of the same SDS-PAGE gel of the pellets following avidin precipitation $\pm$ EDC cross-linking (as indicated) of 93 nM biotinylated tau and/or 3 nM ^{35}S -labeled tau or luciferase (as indicated) bound to $20 \mu\text{M}$ microtubules ($\sim 200:1$ molar ratio of tubulin dimers to tau monomers). The arrow indicates directly cross-linked biotinylated and radiolabeled tau dimers.

negligible amounts of tubulin pellet nonspecifically with the biotinylated tau (Fig. 2A, lane 6).

Finally, to determine whether MT bound tau might homooligomerize, we conducted the experiment using radiolabeled and biotinylated tau at a tau:tubulin molar ratio of $\sim 1:200$. Both EDC cross-linked and non-cross-linked samples were subjected to avidin affinity precipitation, and the resulting pellets were fractionated by SDS-PAGE. Most importantly, radiolabeled tau migrating in the region of the gel predicted for tau-tau dimers (~ 130 kDa) is observed only when radiolabeled tau, EDC, and MTs are present in the reaction (Fig. 2B, compare lanes 9–12). Furthermore, co-migration of biotinylated tau and radiolabeled tau at ~ 130 kDa strengthens the conclusion that the band corresponds to a tau-tau dimer (Fig. 2B, lanes 6 and 12). Specificity of the biotinylated tau for radiolabeled tau is demonstrated by the failure of radiolabeled luciferase to cross-link with biotinylated tau (Fig. 2B, lanes 7 and 8). Taken together, the simplest interpretation for the data is that MT binding leads to intermolecular tau-tau association, even at these very low molar ratios of tau to tubulin.

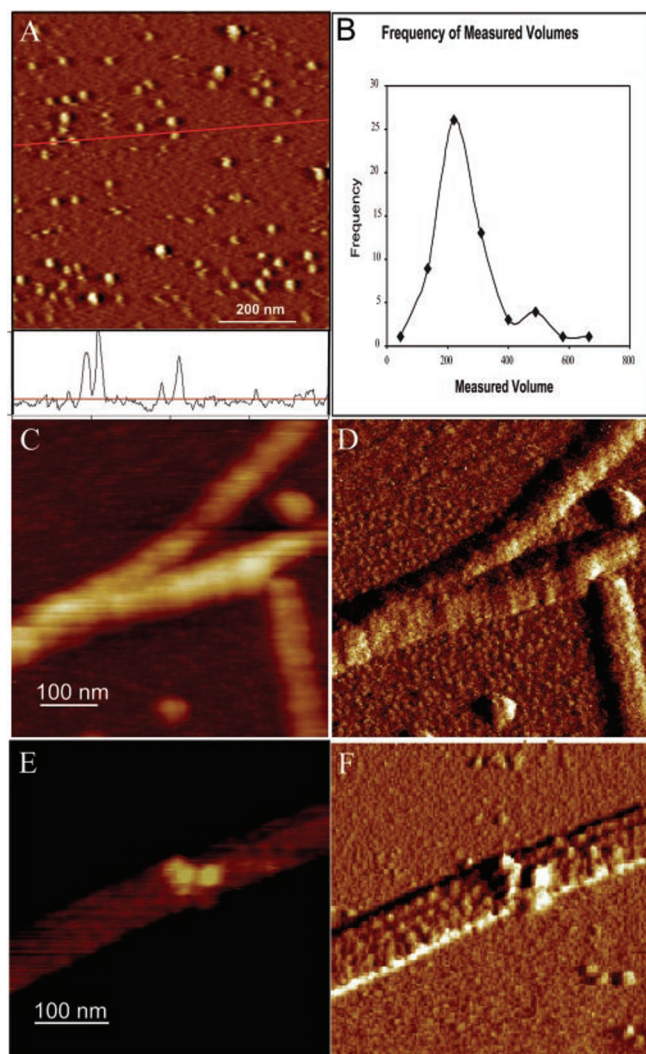


FIG. 3. AFM imaging of the surface structure of highly purified tau, *in vitro* polymerized tubulin, and of tau bound to MTs at saturating ratios of tau-tubulin. For all panels AFM was performed in fluid using intermittent contact. For *C–F*, the topography data from the imaged sample are displayed on the *left*, and the amplitude image is displayed on the *right*. *A*, AFM imaging of highly purified tau reveals thin, disk-like structures. Height data of HPLC-purified reduced tau (5 ng/ml) deposited on mica in BRB-80 (80 mM pipes, 1 mM MgSO_4 , 1 mM EGTA) and imaged in fluid at a 1- μm scan size are shown. A representative section (indicated by the red line on the micrograph) is shown at the bottom of the image. *B*, the frequency of the measured volumes indicates the presence of two populations of tau subunits. The protein volume was calculated (see “Materials and Methods”) from the measured height and the width at half the maximal height for 57 discrete proteins shown in *A*. *C* and *D*, AFM imaging of *in vitro* assembled MTs reveals individual subunits. Prior to imaging 20 μM MTs in BRB-80 stabilized with 10 μM taxol were fixed in 0.2% glutaraldehyde, absorbed onto freshly prepared 3-aminopropyltriethoxysilane-coated mica for 15 s, rinsed, and scanned in sterile filtered water. A 567-nm scan is shown. *E* and *F*, at a high tau-tubulin ratio, tau completely covers the MT surface. Shown is a 460-nm scan of 20 μM taxol-stabilized MTs bound with 10 μM tau. The sample was prepared for imaging as indicated in *C*.

AFM Imaging of Highly Purified Tau Reveals Thin, Disk-like Structures—As an independent route to assess the tau-tau and tau-MT interactions, we next employed tapping mode atomic force microscopy. Using this modality, samples can be imaged in fluid, which allows the hydrated topographical conformation to be observed. We first examined highly purified recombinant 4-repeat tau in the absence of microtubules (Fig. 3*A*). Analysis of the volume distribution ($n = 57$; see formula under “Materials and Methods”) revealed the presence of two peaks (Fig.

3*B*). The measured mean width and height, and the calculated mean molecular volume, were found to be 18.6 ± 2.3 nm, 1.27 ± 0.3 nm, and 174 nm³ for the major peak and 24.2 ± 2.2 nm, 1.95 ± 0.3 nm, and 453.5 nm³ for the minor peak. Thus, under these conditions, tau possesses a thin disk-like structure. The major peak contained 85% of the measured tau subunits ($n = 49$). The simplest interpretation is that the majority of the tau imaged was individual monomers, while only a small proportion ($\sim 15\%$) was dimeric.

AFM Imaging of *in Vitro* Assembled Microtubules Reveals Individual Subunits—Next, we imaged highly purified tubulin that had been assembled into polymers in the presence of taxol. Fig. 3, *C* and *D*, present images of two bundled MTs bifurcating into individual MTs. The image in Fig. 3*C* shows the topography (height) data; the image in Fig. 3*D* is the amplitude (error) image for the same sample. From the topography data, we determined that the MT bundle is ~ 87 nm in diameter and 20 nm in height. The separated individual MTs are ~ 45 – 47 nm in diameter and have a height of 15.2 and 15.4 nm. The lower than expected height (relative to dimensions observed by electron microscopy (33)) is likely due to compression of the MTs resulting from tip-sample interactions (34–36). The larger than expected width is likely due to both compression from tip-sample interactions as well as tip broadening effects inherent in AFM imaging (32). In the amplitude image (Fig. 3*D*), we observed a highly regular array of very small subunits. Since these features are near in size to the sensitivity limit of the AFM feedback mechanism, it would require excessive image processing to enhance these features in the topography image. On the other hand, the small amplitude changes persist in the error mode image. Our interpretation is that this subunit structure reflects the organization of tubulin dimers into longitudinal protofilaments. Given that the MT has a width of ~ 46 nm, and that 6 rows of protofilaments are observed, the apparent subunit diameter is ~ 7.7 nm. This value is in reasonable agreement with dimensions as assessed by electron crystallography (37).

At a High Tau:Tubulin Ratio, Tau Saturates the Microtubule Surface—Fig. 3, *E* and *F* display a MT that has been incubated with saturating levels of tau (~ 1 tau per 2 tubulin dimers). The imaged structure is 79 nm in width and 11.1 nm in height. The surface is covered with four rows of subunits. Dividing the width of the structure by the number of rows yields an effective diameter of ~ 20 nm per subunit, which is two to three times the size of the tubulin dimers observed in the absence of tau in Fig. 3, *C* and *D*, but in good agreement with the dimensions we observed earlier for monomeric tau (Fig. 3*A*), leading to the conclusion that the entire surface of the MT is coated with tau. Occasional secondary layers of clustered structures were also observed (Fig. 3, *E* and *F*). The measured diameter of the clustered subunits in the secondary layer is ~ 25 – 30 nm, and the height may be as much as 5–10 nm, suggesting that these structures may be oligomers composed of stacked tau monomers.

At a Low Tau:Tubulin Molar Ratio, Tau Forms Ring-like Oligomers—As a follow up to the biochemical approaches used in Figs. 1 and 2, we next used atomic force microscopy to independently assess whether MT binding might influence tau-tau associations. We imaged MTs bound with a low (1:200) molar ratio of tau to tubulin dimers (Figs. 4, *A* and *B*). Given the dimensions previously measured for the unbound tau and undecorated MTs, this molar ratio of tau:tubulin should result in vastly sub-saturating coverage of the MT surface. Under these conditions, we observed that the regular surface of the MT was sometimes modified to include structures that were not seen in the absence of bound tau. These tau-dependent

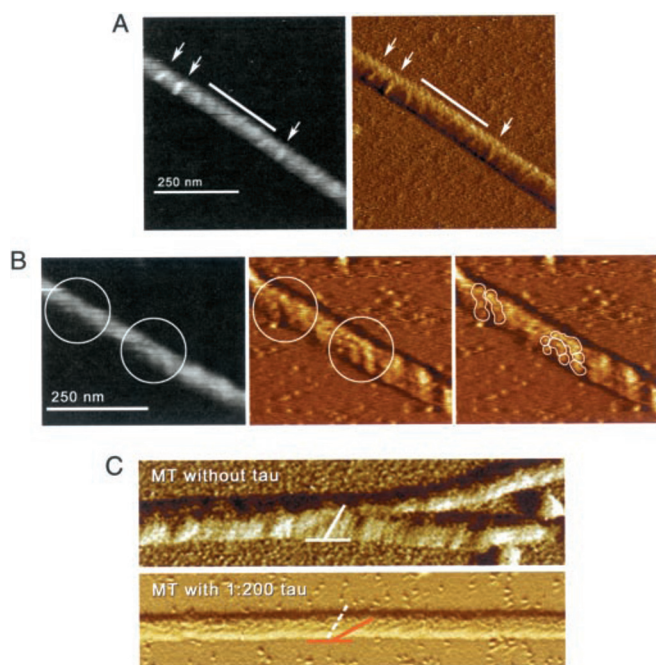


FIG. 4. Surface structure of MTs bound with a low molar ratio of tau reveals the presence of tau oligomers and a possible shift in the relative orientation of microtubule surface features. For all panels 0.2% glutaraldehyde fixed MTs (20 μM) bound with 0.1 μM tau were scanned in fluid as described previously in Fig. 3C. **A**, at a low tau-tubulin molar ratio, tau forms ring-like oligomers. Shown is a 750-nm section of a 1.3- μm scan in which several putative tau oligomers (indicated by arrows in the topography data, left panel, and the amplitude image, right panel) are observed adjacent to an undecorated MT region (indicated by the line). The data for Table I were obtained from the undecorated (MT, line) or putative tau oligomer (arrows) regions of the topography image. **B**, globular subunits and ring-like structures are closely associated on a region of a MT bound with a low level of tau. 509-nm scan, putative tau subunits and oligomers are circled (left and center panels) or outlined (right panel). All panels show the topography data. The image in the center and right panels has been processed using a high pass gaussian filter to emphasize the surface feature details. **C**, tau binding may result in a MT conformational change. Both panels are amplitude images. The top panel is a ~ 955 nm section of a 1.074- μm scan of the MT shown in Fig. 3D. Orientation of the surface features relative to the long axis of the MT is indicated by the white line ($\sim 60^\circ$). The bottom panel shows a 978.5-nm scan of an undecorated region of a MT bound with a low ratio of tau in which the orientation of the surface features is indicated by the red line ($\sim 30^\circ$). For comparison the orientation of the surface structures from the MT shown in the upper panel is indicated by the dotted white line.

structures can be grouped into two general types.

The first group appears as periodic, ring-like structures which span the girth of the MT approximately perpendicular to the long axis (Fig. 4A, arrows). To test the plausibility of the model that the rings are a single layer of tau oligomers encircling the MT, the predicted circumference of such a tau layer (based on the dimensions of tau monomers observed in Fig. 3A) was compared with the observed dimensions. Section analysis was used to measure the average height and width of three representative tau-dependent features (arrows) and of three random areas from the adjacent region of the MT, which lacked the tau-dependent structures (marked with a line). As seen in Table I, the samples from the undecorated MT region had a mean circumference of 117.8 ± 0.1 nm (mean height of 9.1 ± 0.2 nm and mean width of 55.9 ± 0 nm, $n = 3$), while the immediately adjacent putative tau oligomeric structures had a mean circumference of 122.1 ± 1.1 nm (mean height of 10.7 ± 0.8 nm and mean width of 57.7 ± 0.8 nm, $n = 3$). There is a less than 5% probability that these two regions of the MT have the same dimensions (see Table I). Addition of the mean height of

TABLE I

Height, width, and calculated circumferences of undecorated and putative tau oligomer containing regions of a MT

All dimensions were calculated from the topography data using the average section analysis tool. MT data was measured by sampling three random regions from the area in Fig. 4A indicated by the line; MT + tau data measured from the three putative tau oligomers indicated by arrows in Fig. 4A. The probability that each value was derived from the same region was calculated using the t test.

MT region	Height	Width	Circumference
	nm	nm	
MT	9.012	55.92	117.8
MT	8.923	55.92	117.7
MT	9.295	55.92	117.9
Mean MT	9.08	55.92	117.8 ^{a,c}
S.D.	0.19	0.0	0.08
MT + tau	11.23	57.23	121.5
MT + tau	11.00	55.23	121.4
MT + tau	9.75	58.54	123.4
Mean MT + tau	10.66	57.00	122.1 ^{a,b}
S.D.	0.80	1.67	1.12
Calculated MT + tau	11.62	58.47	124.2 ^{b,c}

^a $p < 0.05$.

^b $p < 0.001$.

^c $p < 0.1$, not statistically significant.

tau monomers (1.27 nm) to the dimensions of the undecorated region of the MT was used to model a ring of tau as follows: 11.6 nm in height, 58.5 nm in width, yielding a circumference of 124.2 nm. This calculated circumference is consistent with the measured values for the region of the MT containing the putative tau oligomers ($p > 0.1$) but not for the naive region of the MT ($p \ll 0.001$).

The second type of tau-dependent structures observed were clusters of less organized subunits. Fig. 4B shows a region of a MT possessing a number of such structures, some of which are assembled into a linear structure traversing the MT. The dimensions of these structures are consistent in size with tau monomers as determined earlier.

Finally, in addition to tau-dependant structures, we often observed apparent tau binding dependent changes in MT surface features. The top panel of Fig. 4C is a larger area scan of the MT previously shown in Fig. 3D, while the bottom panel shows a representative region of a MT bound with a low molar ratio of tau that contained no obvious tau structures. In both cases, the surface of the MTs is organized into regular rows. As seen in Fig. 4C, these rows are often observed to converge either along the surface joining two MTs (top panel) or possibly at seams (bottom panel), producing a barbed appearance. Furthermore, in non-tau-containing MTs, the rows were observed to be oriented at an $\sim 60^\circ$ angle to the MT long axis (indicated by the white line, top panel). In contrast in regions of MTs bound with low levels of tau that contained no demonstrable tau structures, the surface orientation of the MT substructure appears to be shifted $\sim 30^\circ$ such that the rows are no longer oriented as nearly perpendicular to the MT long axis (red line versus dotted white line).

DISCUSSION

In this study, we have presented two independent lines of evidence supporting the conclusion that tau oligomerizes upon binding to MTs. First, co-precipitation experiments demonstrated that tau forms tau-tau complexes in a MT-dependent manner, even at very low molar ratios of tau to tubulin. Second, atomic force microscopic images of microtubules bound with low levels of tau revealed tau oligomers in ring-like structures encircling the microtubules. Furthermore, at saturating tau to tubulin ratios, tau covered the entire MT surface in a regular

structural pattern. The formation of tau oligomers may be an important element in the ability of tau to interact productively with microtubules as well as have mechanistic implications for its ability to regulate their dynamic behavior.

One question that arises is whether or not the generation of tau oligomers might result from the cross linker “catching” individual tau monomers contacting one another as they slide along the microtubule surface. While this is a formal possibility, we do not believe this is likely based on several arguments. First, using a very low ratio of tau to tubulin (1:200), we found the rate of tau-tau cross-linking by EDC to be extremely rapid, with observable cross-linked dimers forming in less than 10 s (data not shown). Second, the identification of brain MAPs was originally based on the strong affinity that MAPs exhibit for microtubules during multiple cycles of polymerization and depolymerization (38). Importantly, Weingarten *et al.* (38) demonstrated that tau remained associated with the identical protofilament fraction during repeated cycles of polymerization and de-polymerization, *i.e.* no tau exchange between tubulin subunits was observed. Most directly, Job *et al.* (39) developed a biochemical assay that distinguished between MAP and non-MAP containing regions of microtubules. Using this assay they showed that once MAPs are bound they do not exchange between or migrate along MTs (39). Thus, we conclude that the oligomers we observed are not the result of tau migration or exchange. Rather, the simplest conclusion is that cross-linking detects intermolecular tau-tau associations by covalently coupling pre-existing non-covalent tau oligomers.

Tau Structures Observed by AFM—Among the strengths of AFM is the ability to image individual hydrated protein molecules and multimolecular assemblies in an aqueous environment at relatively high resolution. Using these conditions, we observed tau 1) as purified tau, adsorbed onto mica, 2) bound to MTs at a saturating molar ratio of tau to tubulin, and 3) at a very low molar ratio of tau to tubulin.

A variety of previous studies have led to the conclusion that purified tau in solution has an extended, predominantly random coil structure with little secondary or tertiary structure (40–42). In contrast, when adsorbed to mica, we found that putative tau monomers possessed a disk-like structure with a diameter of 18.5 nm and a height of 1.3 nm. Putative tau dimers exhibited only a 1.3-fold increase in length and a 1.5-fold increase in height, suggesting a closely associated tau-tau dimeric structure.

In assessing the degree of folding present in these tau monomers, the following considerations are useful. First, an entirely unfolded protein of tau’s size (432 residues) would be expected to have a length of ~145 nm and a diameter of less than 1 nm (43). At the other extreme, a compact globular protein of similar molecular mass such as actin (45 kDa) exhibits dimensions of 13.9 nm in width and in 1.1 nm height using analogous AFM methodology (44). Based on these numbers, we conclude that tau is neither fully extended nor tightly folded under our imaging conditions. This raises the question of why tau appears to have a different structure on mica than in solution. One possible explanation could be that tau’s association with the negatively charged mica partially mimics its binding to the highly acidic C terminus of tubulin in microtubules. Consistent with this possibility, subunits that were approximately the size of tau monomers (based on the data in Fig. 3, *A* and *B*) covered the surface of microtubules incubated with saturating concentrations tau (Fig. 3, *E* and *F*).

At very low tau to tubulin ratios, we observed two putative structures for microtubule bound tau. One appeared as narrow, raised structures that wrapped around the microtubule in a ring-like fashion (Fig. 4*A*). The calculations in Table I demon-

strate that the dimensions of these rings are consistent with them being composed of a single layer of tau subunits. The second MT-bound tau structure appeared as clustered disk-like subunits, again with the approximate dimensions of tau monomers (Fig. 4*B*). It is possible that the less ordered clusters are in the process of assembling into more ordered ring like structures.

How Might Tau Stabilize Microtubule Dynamics—One of the current enigmas in the field is the ability of tau to regulate microtubule dynamics when present at very low levels (12). Based on these observations, it has been proposed that tau binding to microtubules induces an allosteric wave in the microtubule resulting in altered dynamic behavior. Here, we see rings of tau encircling the entire microtubule; thus, tau can have direct structural impact on every protofilament. Taken together with the allosteric wave model, the net result could be tau-induced structural changes extending outwards from the tau ring causing altered dynamic behavior. Comparison of the images in Fig. 4*C* are consistent with this possibility.

The nature of the structural changes shown in Fig. 4*C* may be indicative of changes in the protofilament orientation. However, such large changes seem unlikely. Alternatively, it has been proposed that MAP binding may propagate an ordered conformation of the disordered C-terminal tail of tubulin, thereby stabilizing the polymer form.³ Tuszyński and co-workers³ have modeled the interactions of the highly electronegative C termini of tubulin with adjacent bound MAPs. They found that C termini are likely to exist in several conformational states, including a state in which the tubulin C termini bind to the MT surface in a positively charged groove. This orientation is predicted to exhibit a regular arrangement of binding sites and therefore to form a fairly regular lattice structure. This regularly ordered condensed C termini structure may be detectable by AFM and provide the basis for the appearance of a “shift” in the tau-bound MT surface that we observed.

In addition to the above described allosteric mechanism, it is also possible that tau rings may strengthen lateral protofilament-protofilament associations. In this regard, it is also important to note recent cryo-electron microscopic analyses of the tau-microtubule interaction, suggesting that tau stabilizes longitudinal interactions along individual protofilaments (45). Given the very different technologies employed in our work and that of Al-Bassam *et al.* (45), at this point, there is no reason to view these models as mutually exclusive. More data will be necessary to assess each of them.

Molecular Mechanisms by Which Tauopathies May Result in Neuronal Cell Death—Our results lead us to the hypothesis that oligomerization may be required for normal tau function. This possibility also has important mechanistic implications for molecular mechanisms underlying tau-mediated cell death in FTDP-17; it could be equally relevant mechanistically in the other tauopathies, including Alzheimer’s disease.

As noted in the introduction, the most common molecular mechanism underlying dominant mutations is poisoning or otherwise altering the activity of an oligomeric complex. A rarer but still very important mechanism leading to a dominant mutant phenotype is for the mutated protein to gain a toxic function. Tau oligomerization could play a key role in either mechanism. For example, several authors have suggested that the FTDP-17 tau mutations causing amino acid substitutions lead to an increased probability of forming abnormal tau oligomers and then fibers (paired helical filaments/neurofibrillary tangles) (46, 47). Abnormal oligomerization, the

³ J. A. Tuszyński, personal communication.

gained function, could be a precursor state to assembly of the larger cytotoxic fibers.

On the other hand, it is well established that microtubule dynamics must be tightly controlled by cells. Indeed, relatively small changes in the dynamic behavior of microtubules, for example by taxol, lead to apoptosis (14). We have recently proposed a haplo-insufficiency model in which tau-mediated regulation of microtubule dynamics must maintain the level of dynamic behavior within a narrow range of activities to maintain viability; over- or under-dynamic microtubules lead to cell death.⁴ Taken together with the data presented here, we propose that the formation of tau oligomers containing FTDP-17 amino acid substitution mutants reduces the ability of tau to stabilize microtubule dynamics (*i.e.* loss-of-function), thereby causing cell death. With respect to the FTDP-17 tau RNA splicing mutations, there are multiple possibilities. If tau can form only homo-oligomers (4-repeat tau oligomerizes with only 4-repeat tau and 3-repeat tau only with 3-repeat tau, each with distinct properties), then the splicing mutations could cause cell death by having an excess of 4-repeat homo-oligomers overstabilizing microtubule dynamics. Alternatively, if mixed oligomers can form, then oligomers of different compositions would be expected to possess different functional and mechanistic capabilities.⁴ Again, the whole population of tau oligomers in the splicing mutations would have an excess of 4-repeat tau, leading to overstabilization of microtubules. Consistent with the loss-of-function model, all but one of the dozen or so FTDP-17 mutations that cause amino acid substitutions map to regions of the protein known to either be involved in microtubule binding or to regulate microtubule binding; the one exception may be involved in tau function via tertiary structure.

Acknowledgments—We are enormously thankful to Herb Miller, Les Wilson, and MaryAnn Jordan for insightful suggestions and generous contributions of tubulin. We also thank Monte Radeke for several valuable conversations and Adam Mechler for helpful comments on the manuscript.

REFERENCES

- Esmaeli-Azad, B., McCarty, J. H., and Feinstein, S. C. (1994) *J. Cell Sci.* **107**, 869–879
- Knops, J., Kosik, K. S., Lee, G., Pardee, J. D., Cohen-Gould, L., and McConlogue, L. (1991) *J. Cell Biol.* **114**, 725–733
- Liu, C. W., Lee, G., and Jay, D. G. (1999) *Cell Motil. Cytoskeleton* **43**, 232–242
- Takei, Y., Teng, J., Harada, A., and Hirokawa, N. (2000) *J. Cell Biol.* **150**, 989–1000
- Dawson, H. N., Ferreira, A., Eyster, M. V., Ghoshal, N., Binder, L. I., and Vitek, M. P. (2001) *J. Cell Sci.* **114**, 1179–1187
- Kanai, Y., Takemura, R., Oshima, T., Mori, H., Ihara, Y., Yanagisawa, M., Masaki, T., and Hirokawa, N. (1989) *J. Cell Biol.* **109**, 1173–1184
- LoPresti, P., Szuchet, S., Papasozomenos, S. C., Zinkowski, R. P., and Binder, L. I. (1995) *Proc. Natl. Acad. Sci. U. S. A.* **92**, 10369–10373
- Goedert, M., Spillantini, M. G., Crowther, R. A., Chen, S. G., Parchi, P., Tabaton, M., Lanska, D. J., Markesbery, W. R., Wilhelmsen, K. C., Dickson, D. W., Peterson, R. B., and Gambetti, P. (1999) *Nat. Med.* **5**, 454–457
- Fellous, A., Francon, J., Lennon, A. M., and Nunez, J. (1977) *Eur. J. Biochem.* **78**, 167–174
- Drechsel, D. N., Hyman, A. A., Cobb, M. H., and Kirschner, M. W. (1992) *Mol. Biol. Cell* **3**, 1141–1154
- Trinczek, B., Biernat, J., Baumann, K., Mandelkow, E. M., and Mandelkow, E. (1995) *Mol. Biol. Cell* **6**, 1887–1902
- Panda, D., Daijo, J. E., Jordan, M. A., and Wilson, L. (1995) *Biochemistry* **34**, 9921–9929
- Goncalves, A., Braguer, D., Kamath, K., Martello, L., Briand, C., Horwitz, S., Wilson, L., and Jordan, M. A. (2001) *Proc. Natl. Acad. Sci. U. S. A.* **98**, 11737–11742
- Yvon, A. M., Wadsworth, P., and Jordan, M. A. (1999) *Mol. Biol. Cell* **10**, 947–959
- Lee, V. M., Goedert, M., and Trojanowski, J. Q. (2001) *Annu. Rev. Neurosci.* **24**, 1121–1159
- Hutton, M., Lendon, C. L., Rizzu, P., Baker, M., Froelich, S., Houlden, H., Pickering-Brown, S., Chakraverty, S., Isaacs, A., Grover, A., Hackett, J., Adamson, J., Lincoln, S., Dickson, D., Davies, P., Petersen, R. C., Stevens, M., de Graaff, E., Wauters, E., van Baren, J., Hillebrand, M., Joosse, M., Kwon, J. M., Nowotny, P., Heutink, P., et al. (1998) *Nature* **393**, 702–705
- Spillantini, M. G., Murrell, J. R., Goedert, M., Farlow, M. R., Klug, A., and Ghetti, B. (1998) *Proc. Natl. Acad. Sci. U. S. A.* **95**, 7737–7741
- Clark, L. N., Poorkaj, P., Wszolek, Z., Geschwind, D. H., Nasreddine, Z. S., Miller, B., Li, D., Payami, H., Awert, F., Markopoulou, K., Andreadis, A., D'Souza, I., Lee, V. M., Reed, L., Trojanowski, J. Q., Zhukareva, V., Bird, T., Schellenberg, G., and Wilhelmsen, K. C. (1998) *Proc. Natl. Acad. Sci. U. S. A.* **95**, 13103–13107
- Hong, M., Zhukareva, V., Vogelsberg-Ragaglia, V., Wszolek, Z., Reed, L., Miller, B. I., Geschwind, D. H., Bird, T. D., McKeel, D., Goate, A., Morris, J. C., Wilhelmsen, K. C., Schellenberg, G. D., Trojanowski, J. Q., and Lee, V. M. (1998) *Science* **282**, 1914–1917
- Himmler, A. (1989) *Mol. Cell. Biol.* **9**, 1389–1396
- Himmler, A., Drechsel, D., Kirschner, M. W., and Martin, D. W., Jr. (1989) *Mol. Cell. Biol.* **9**, 1381–1388
- Butner, K. A., and Kirschner, M. W. (1991) *J. Cell Biol.* **115**, 717–730
- Gustke, N., Trinczek, B., Biernat, J., Mandelkow, E. M., and Mandelkow, E. (1994) *Biochemistry* **33**, 9511–9522
- Goode, B. L., Chau, M., Denis, P. E., and Feinstein, S. C. (2000) *J. Biol. Chem.* **275**, 38182–38189
- Lee, G., Neve, R. L., and Kosik, K. S. (1989) *Neuron* **2**, 1615–1624
- Goedert, M., and Jakes, R. (1990) *EMBO J.* **9**, 4225–4230
- D'Souza, I., Poorkaj, P., Hong, M., Nochlin, D., Lee, V. M., Bird, T. D., and Schellenberg, G. D. (1999) *Proc. Natl. Acad. Sci. U. S. A.* **96**, 5598–5603
- Taylor, J. P., Hardy, J., and Fischbeck, K. H. (2002) *Science* **296**, 1991–1995
- Gamblin, T. C., King, M. E., Dawson, H., Vitek, M. P., Kuret, J., Berry, R. W., and Binder, L. I. (2000) *Biochemistry* **39**, 6136–6144
- Goedert, M., Jakes, R., and Crowther, R. A. (1999) *FEBS Lett.* **450**, 306–311
- Chau, M. F., Radeke, M. J., de Ines, C., Barasoain, I., Kohlstaedt, L. A., and Feinstein, S. C. (1998) *Biochemistry* **37**, 17692–17703
- Fritzsche, W., Vesenska, J., and Henderson, E. (1995) *Scanning Microsc.* **9**, 729–739
- McEwen, B., and Edelman, S. J. (1980) *J. Mol. Biol.* **139**, 123–145
- Lal, R., and John, S. A. (1994) *Am. J. Physiol.* **266**, C1–C21
- Muller, D. J., and Engel, A. (1997) *Biophys. J.* **73**, 1633–1644
- Baumgartner, W., Hinterdorfer, P., and Schindler, H. (2000) *Ultramicroscopy* **82**, 85–95
- Nogales, E., Wolf, S. G., and Downing, K. H. (1998) *Nature* **391**, 199–203
- Weingarten, M. D., Lockwood, A. H., Hwo, S. Y., and Kirschner, M. W. (1975) *Proc. Natl. Acad. Sci. U. S. A.* **72**, 1858–1862
- Job, D., Pabion, M., and Margolis, R. L. (1985) *J. Cell Biol.* **101**, 1680–1689
- Cleveland, D. W., Hwo, S. Y., and Kirschner, M. W. (1977) *J. Mol. Biol.* **116**, 227–247
- Wille, H., Drewes, G., Biernat, J., Mandelkow, E. M., and Mandelkow, E. (1992) *J. Cell Biol.* **118**, 573–584
- Schweers, O., Schonbrunn-Hanebeck, E., Marx, A., and Mandelkow, E. (1994) *J. Biol. Chem.* **269**, 24290–24297
- Alberts, B., Bray, D., Lewis, J., Raff, M., Roberts, K., and Watson, J. D. (1989) *Molecular Biology of the Cell*, Second Ed., p. 116, Garland Publishing, Inc., New York
- Schneider, S. W., Larmer, J., Henderson, R. M., and Oberleithner, H. (1998) *Pfluegers Arch.* **435**, 362–367
- Al-Bassam, J., Ozer, R. S., Safer, D., Halpain, S., and Milligan, R. A. (2002) *J. Cell Biol.* **157**, 1187–1196
- Li, L., von Bergen, M., Mandelkow, E. M., and Mandelkow, E. (2002) *J. Biol. Chem.* **277**, 41390–41400
- Tatebayashi, Y., Miyasaka, T., Chui, D. H., Akagi, T., Mishima, K., Iwasaki, K., Fujiwara, M., Tanemura, K., Murayama, M., Ishiguro, K., Planel, E., Sato, S., Hashikawa, T., and Takashima, A. (2002) *Proc. Natl. Acad. Sci. U. S. A.* **99**, 13896–13901

⁴ Panda, D., Samuel, J. C., Massie, M., Feinstein, S. C., and Wilson, L. (2003) *Proc. Natl. Acad. Sci. U. S. A.*, in press.